

ASM Notes (L-3)

One-Sample Estimation & Hypothesis Testing

Context

- One sample center.
- We have covered hypothesis testing. Now we move to **Estimation**.

One-Sample Estimation

Let θ = population median.

- One point estimate is the sample median $\hat{\theta}$.

Interval Estimation

We derive this by **inverting the hypothesis test**. We know from general inference that this can be done.

Derivation of Interval Estimates

Model Setup

Let θ be the population median (population median).

$$X_1, \dots, X_n \stackrel{iid}{\sim} F, \quad \text{med}(F) = \theta$$

Hypothesis and Test Statistic

Given the null hypothesis:

$$H_0 : \theta = a$$

Then the test statistic Y is defined as:

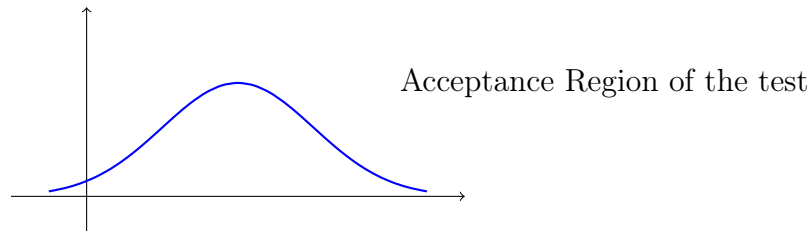
$$Y = \#\{X_i < a\} \sim \text{Bin}\left(n, \frac{1}{2}\right) \quad \text{under } H_0.$$

Acceptance Region

The bounds for the acceptance region are determined by the quantiles of the Binomial distribution:

$$\text{qbinom}\left(\frac{\alpha}{2}, n, \frac{1}{2}\right) < Y < \text{qbinom}\left(1 - \frac{\alpha}{2}, n, \frac{1}{2}\right)$$

Observed Acceptance Region:



The observed acceptance region is specified as:

$$179 < Y < 218$$

Confidence Interval Construction

Using the count from the data:

$$\#(X_i < a) = 179$$

We invert the test to find the bounds for the parameter a :

$$a = X_{(179)}$$

Thus, the Confidence Interval (CI) is:

$$CI = (X_{(179)}, X_{(218)})$$

Two-Sample Test for Center

Confidence Interval Conclusion: $(X_{(L)}, X_{(U)})$ is the CI for a where $(L < Y < U)$ is the acceptance region of the test.

Two-Sample Test Setup

Let $X_1, \dots, X_{n_1} \stackrel{iid}{\sim} D_1$ and $Y_1, \dots, Y_{n_2} \stackrel{iid}{\sim} D_2$.

Hypothesis:

$H_0 : D_1 \equiv D_2$ (Distributions are identical)

$H_a : D_1$ is shifted to the right of D_2 (Right-tailed) or

$H_a : D_1$ is shifted to the left of D_2 (left-tailed) or

$H_a : D_1$ is either shifted to the right or shifted to the left of D_2 (two-sided)

Assumptions:

- Continuous distributions.
- They have the same shape (shift model).
- Independence of data values.

Wilcoxin Rank Sum Test**Procedure:**

1. Pool together X 's and Y 's.
2. Order the pooled sample.
3. Sum the ranks of X 's in the pooled sample.

Logic:

- If X 's are larger than Y 's (Right-tailed alternative), then D_1 is shifted to the right of D_2 .
- Consequently, the average ranks of X 's will be larger than the average ranks of Y 's.

Test Statistic (T): Here we focus on the rank sum T .

T = is the sum of ranks of X_i 's in the pooled sample $-\frac{(n_1)(n_1 + 1)}{2}$

$$T + T_1 = \frac{(n_1 + n_2)(n_1 + n_2 + 1)}{2} - \frac{(n_1)(n_1 + 1)}{2} = C$$

T_1 = the sum of ranks of Y_i 's.

Numerical Example**Relationship between Rank Sums:**

$$T_1 = C - T$$

Example: Reaction Time of Subjects

We compare reaction times under influence of two drugs (or treatments).

Data:

$$X = (2.07, 2.11, 2.43, 2.50, 2.71, 2.84, 2.88) \quad (n_1 = 7)$$

$$Y = (1.62, 1.71, 1.93, 1.96, 2.24, 2.41) \quad (n_2 = 6)$$

Calculation of Ranks: We pool and sort the data to assign ranks. The ranks of X_i in the pooled sample are:

$$\text{Ranks of } X_i = (5, 6, 9, 10, 11, 12, 13)$$

Verification:

- Y 's occupy ranks: 1, 2, 3, 4, 7, 8.
- X 's occupy ranks: 5, 6, 9, 10, 11, 12, 13.

Statistic Calculation:

$$\text{Sum of Ranks of } X(T) = 5 + 6 + 9 + 10 + 11 + 12 + 13 = 66$$

To calculate the Mann-Whitney U statistic (subtracting minimum rank sum):

$$U = 66 - \frac{n_1(n_1 + 1)}{2} = 66 - \frac{7(8)}{2} = 66 - 28 = 38$$

Result:

- P-value = 0.01399 (two-tailed)
- At $\alpha = 5\%$, we reject H_0 , i.e. $D_1 \neq D_2$.

Null Distribution Construction

How do we find the distribution of the test statistic under H_0 ($D_1 \equiv D_2$)?

1. Generate all combinations of size n_1 from the pooled set of size $n_1 + n_2$.
2. Total number of combinations: $\binom{n_1 + n_2}{n_1}$.
3. For each such combination, label them as "Group X" and calculate T .
4. Since H_0 implies exchangeability, all combinations are equally likely (probability = $1/\binom{n_1 + n_2}{n_1}$).
5. This generates the "Null Distribution".
6. Find the position of the observed T in this distribution to get the p-value.

Permutation Tests**General Concept**

Such tests that use permutations to obtain the null distribution are called **Permutation Tests**.

Application to Parametric Statistics: Even for a parametric statistic, we can use this technique if the distribution assumption is violated.

Example: Building a t-statistic.

$$t = \frac{\bar{x} - \bar{y}}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

- We do not know the distribution of this statistic unless X 's and Y 's are Normal.

- However, we can generate the distribution using the **Permutation Technique**.

Procedure:

1. Shuffle the data labels.
2. Re-calculate the t -statistic for every shuffle.
3. This empirical distribution serves as the null distribution, allowing us to perform the test without assuming Normality.

Week-II

```
# Wilcoxon rank sum test
# Example: Hollander & Wolfe (1973)
# Permeability constants of the human chorioamnion (a placental membrane) at
# term (x)
# and between 12 to 26 weeks gestational age (y). The alternative of interest
# is greater permeability of the human chorioamnion for the term pregnancy.

x <- c(0.80, 0.83, 1.89, 1.04, 1.45, 1.38, 1.91, 1.64, 0.73, 1.46)
y <- c(1.15, 0.88, 0.90, 0.74, 1.21)
z<-sort(c(x,y))
T<-sum(which(z %in% x))-10*11/2
T

ranksum<-function(n,m,T){
  T=T+m*(m+1)/2
  x<-c(1:n)
  l<-choose(n,m)
  k<-rep(0,l)
  y<-combn(x,m)
  for(j in 1:l){
    k[j]<-sum(x[y[,j]])}
  pval<-2*length(which(k<T))/choose(n,m)
  pval<-ifelse(pval>1, 2-pval, pval)
  return(pval)
}
ranksum(15,10,35)

wilcox.test( c(0.80, 0.83, 1.89, 1.04, 1.45, 1.38, 1.91, 1.64, 0.73, 1.46),
c(1.15, 0.88, 0.90, 0.74, 1.21))

# Permutation test using t-statistic
T<-t.test(x,y)$statistic
t.pval<-function(a,b,T){
  c<-cbind(a,b)
  m<-length(b)
  n<-length(c)
  x<-c(1:n)
```

```
l<-choose(n,m)
t<-rep(0,l)
y<-combn(x,m)
for(j in 1:l){
  t[j]<-t.test(c[y[,j]],c[-y[,j]])$statistic}
pval<-2*length(which(t>=T))/l
pval<-ifelse(pval>1, 2-pval, pval)
return(pval)
}
t.pval(x,y,T)
```